Replication of "Measuring Economic Growth from Outer Space"

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Code and replication materials (Github)

Introduction

GDP is among the most important measures of economic performance, yet for much of the world it is measured with considerable error. Official statistics are also rarely available below the national level, which limits what can be learned about growth within countries. Henderson, Storeygard, and Weil (2012a) proposed using satellite measurements of night lights as an independent measure of economic activity. Because the errors in lights data have no reason to be related to the errors made by national statistical agencies, the two measures can be combined into an estimate of true income growth that is more precise than either one alone. Applying this framework, they find that the elasticity of measured GDP with respect to lights is roughly 0.3, that official GDP in countries with weak statistical systems is substantially noisier than elsewhere, and that for those countries official figures and lights deserve roughly equal weight.

In this paper, I reproduce the main analyses of Henderson, Storeygard, and Weil (2012a). My analyses are implemented in the R programming language (R Core Team 2026), and the code is available on Github.[1] The pipeline is structured to facilitate extending the analyses to periods outside the original study's sample, although actually doing so is outside the scope of this project.

The original results reproduce closely. The baseline elasticity of GDP with respect to lights is 0.281, compared with 0.277 in the original paper. The tests for differences between countries with good and bad statistical systems have the same signs, magnitudes, and significance as in the original, and the composite growth estimates for the thirty countries with the weakest statistics are within about 0.06 percentage points per year of the published values. The variables based on unweighted counts of pixels, namely the number of unlit pixels and the spatial Gini coefficient, differ noticeably from the original, because they are sensitive to how water and coastal pixels are assigned to countries. Additionally, the published calibration of the measurement-error model is consistent with a formula for the variance of noise in lights that differs from the one implied by the paper's equations. The difference is small at the authors' preferred parameter values but grows as the assumed quality of official data declines.

[1] https://github.com/gideongrinberg/measuring-growth-from-space

Methods and Theoretical Framework

Henderson, Storeygard, and Weil (hereinafter, "HSW") treat true income Y as unobserved. Let lowercase letters denote logs, so that $y_{j,t} = \ln Y_{j,t}$ for country j in year t. Measured GDP equals true income plus error,

$$z_{j,t} = y_{j,t} + \varepsilon_{z,j,t}.$$

HSW also assume a constant-elasticity relationship between lights X and true income, $X = Y^{\beta}$, where β is the elasticity of lights with respect to income. Taking logs and allowing for measurement error in lights gives

$$x_{j,t} = \beta y_{j,t} + \varepsilon_{x,j,t}.$$

The errors ε_z and ε_x are uncorrelated with each other and with y. Denote their variances by σ_z^2 and σ_x^2, and the variance of true growth by σ_y^2. For prediction, the paper regresses measured GDP growth on lights growth. The resulting coefficient,

$$\psi = \frac{\operatorname{cov}(x,z)}{\operatorname{var}(x)} = \frac{\beta\sigma_y^2}{\beta^2\sigma_y^2 + \sigma_x^2},$$

is an estimate of the inverse of the elasticity of lights with respect to income, scaled down by the share of the variance in lights growth that reflects true income. It is therefore a biased estimate of $1/\beta$, but it is the best linear predictor of income growth from lights growth, and because the error in measured GDP is uncorrelated with lights, it can be estimated without knowing the true income.

In practice, how bright a country is at a given income level also depends on factors unrelated to income, including population density and differences in satellite sensors between years. Country and year fixed effects absorb these trends, giving the specification

$$\ln \mathrm{GDP}_{jt} = \psi \ln(\mathrm{lights/area})_{jt} + c_j + d_t + e_{jt},$$

where c_j represents country fixed effects and d_t represents year fixed effects. The paper also employs a long differences specification, which regresses change in log GDP between 1992/93 and 2005/06 on the change in log lights.

Given fitted values $\hat{z}_j = \hat{\psi} x_j$, HSW construct a composite estimate of true growth: $\hat{y}_j = \lambda z_j + (1-\lambda)\hat{z}_j$. The weight on measured GDP that minimizes the error of the composite is

$$\lambda^* = \frac{\sigma_x^2\sigma_y^2}{\sigma_z^2(\beta^2\sigma_y^2 + \sigma_x^2) + \sigma_x^2\sigma_y^2}.$$

The difficulty is that λ^* depends on four unknown parameters, while the data provide only three,

$$\operatorname{var}(z) = \sigma_y^2 + \sigma_z^2, \qquad \operatorname{var}(x) = \beta^2\sigma_y^2 + \sigma_x^2, \qquad \operatorname{cov}(x,z) = \beta\sigma_y^2.$$

The authors address this challenge with outside information on data quality. They divide countries into those with good and bad statistical systems using the World Bank's ratings of statistical capacity, assume that both groups share the same relationship between lights and

income but differ in σ_z^2, and assume a value for the share of the variance of measured GDP growth that is true signal among countries with good data,

$$\phi_g = \frac{\sigma_y^2}{\sigma_y^2 + \sigma_{z,g}^2}.$$

To solve the model, HSW fix the value of ϕ_g. The variance of measured growth among good data countries then gives σ_y^2, and since bad data countries are assumed to have the same variance of true growth, the rest of their variance pins down $\sigma_{z,b}^2$. With σ_y^2 known, the covariance of lights and GDP gives β, and σ_x^2 is what remains of the variance of lights. The optimal weights for each group then follow from these parameters.

Data and Construction

The lights data are the Version 4 stable lights composites from the Defense Meteorological Satellite Program's Operational Linescan System (DMSP-OLS), hosted by the Earth Observation Group (EOG) at the Payne Institute for Public Policy (Baugh et al. 2010). Each composite averages the nighttime observations of one satellite over one year into a longitude-latitude grid, where each pixel represents the mean intensity of lights on a 30-arcsecond cell. The intensity is reported as a "digital number" (DN) between 0 and 63. HSW rely on 30 satellite years, but I was unable to locate a public copy of the composite produced by satellite F15 in 2008 and therefore analyze only 29 satellite years.

Because grid size varies with latitude, HSW weight each pixel by its land area, which also accounts for coastal pixels that are partly water. Countries are assigned to pixels using the national boundaries from Version 3 of the Gridded Population of the World (CIESIN 2005), which I rasterize onto the lights grid so that each pixel belongs to the country containing its center. Land area per pixel comes from the Global Rural-Urban Mapping Project, Version 1 (CIESIN 2011). The authors used their own rasterization of a cleaned version of the boundaries and an unpublished alpha release of the land area grid. Although both files are in the author's replication package, neither is available from a public source, and the manual edits to the boundaries rasterization are not documented. These deviations, along with the missing F15 data, are the main source of the differences documented below.

For country j and satellite-year s, lights per area is

$$\frac{\text{lights}}{\text{area}} = \frac{\sum_{p \in j} \text{DN}_{ps}\, a_p}{\sum_{p \in j} a_p},$$

where a_p is the land area of pixel p. As in the original paper, I average this total across the satellites observing each year before dividing by land area and taking logs. Following HSW's procedure, a country-year is excluded if, for any satellite, more than 5 percent of the country's land area has no cloud-free observations. The number of top-coded and unlit pixels are counts of pixels with DN equal to 63 and 0, and the spatial Gini coefficient is the Gini coefficient of pixel DN within the country. All three are computed for each satellite-year and averaged within years. Unlike the lights measure, these variables count pixels without weighting by land area.

Income and electricity data come from the October 2009 vintage of the World Development Indicators. I follow HSW's decision to exclude certain countries from the sample: Bahrain,

Singapore, Hong Kong, and Macau are excluded because a large share of their pixels are top-coded, Equatorial Guinea because nearly all of its lights come from gas flares, and Serbia and Montenegro because their borders changed during the period.

The data quality ratings come from Appendix A of World Bank (2002), which reports, for each country, ten indicators of statistical practice such as the base year of the national accounts, the balance of payments manual in use, and the dates of the most recent population and agricultural censuses. I score each indicator using the rules stated in the appendix and sum the scores to obtain a rating from 0 to 10. As in the original paper, I keep the 122 countries that appear in both of the report's country lists, which drops Puerto Rico, Saudi Arabia, and Tunisia, and classify countries rated 3 or below as having bad data. The rated subsample contains 118 countries in the panel and 113 in long differences, of which 30 have scores of 3 or below.

Results

Table 1 reproduces the baseline results (Table 2 in the original paper). The elasticity of GDP with respect to lights in the baseline specification is 0.281 with a standard error of 0.031, against 0.277 and 0.031 in the original, and the within R^2 is 0.770 against 0.769. A quadratic term in lights is small and insignificant (-0.004, against -0.006), confirming that the relationship is close to log-linear. Controlling for top-coded and unlit pixels leaves the elasticity essentially unchanged at 0.272, and the coefficient on top-coded pixels, 0.011, is marginally significant as in the original (0.012). The coefficient on unlit pixels, -0.006 with a standard error of 0.018, is smaller and less precise than the original -0.012 with a standard error of 0.012, and the coefficient on the spatial Gini is 0.127 against 0.165. Both are insignificant in either case, so the substantive conclusion that country fixed effects absorb differences in the dispersion of lights is unchanged, but these are the two estimates that depart most from the original, for reasons discussed in the next section. The comparison with electricity consumption reproduces within 0.002: on the subsample with electricity data, the elasticity with respect to lights is 0.283, the elasticity with respect to electricity is 0.283, and when both enter, the coefficients are 0.168 and 0.201, against 0.282, 0.283, 0.166, and 0.201.

Table 2 reproduces the original Table 3, which examines annual fluctuations and long-run growth. Adding a linear time trend for each country lowers the elasticity to 0.193 with a standard error of 0.037, against 0.180 and 0.036, so lights continue to explain deviations of GDP from each country's growth path. The original standard errors in this column are not clustered by country, while mine are. Splitting the demeaned lights residuals into positive and negative parts yields coefficients of 0.273 and -0.288, against 0.274 and -0.279. These are nearly equal in absolute value, so lights fall in downturns as much as they rise in expansions and there is no evidence of a ratchet effect. The long-difference estimate of the elasticity between 1992/93 and 2005/06 is 0.321 with an R^2 of 0.276, against 0.320 and 0.279, and adding changes in top-coded and unlit pixels moves it only to 0.304, against 0.302.

Table 3 reproduces the results for the countries rated by the World Bank (Table 4 in the original). In this sample the elasticity is 0.310 with country and year fixed effects, 0.280 with country trends, and 0.328 in long differences, against 0.307, 0.270, and 0.327, and the long-difference constant is 0.365, against 0.365. Panel B tests the two assumptions on

Table 1: Baseline Results, 1992-2008

	ln(GDP)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ln(lights/area)	0.281***	0.270***	0.272***	0.288***	0.283***		0.168***
	(0.031)	(0.035)	(0.032)	(0.034)	(0.047)		(0.051)
ln(lights/area)^2		−0.004					
		(0.006)					
ln(top-coded + 1)			0.011*				
			(0.006)				
ln(unlit + 1)			−0.006				
			(0.018)				
Spatial Gini				0.127			
				(0.212)			
ln(KWH)						0.283***	0.201***
						(0.047)	(0.040)
Observations	3016	3016	3016	3016	1853	1853	1853
Countries	188	188	188	188	128	128	128
(Within) R^2	0.770	0.771	0.772	0.771	0.758	0.767	0.782

* p <0.1, ** p <0.05, *** p <0.01
Clustered (country) standard errors in parentheses.

Table 2: Lights Up/Down, Time Trend, and Long Differences

	Fixed effects	Country time trend	Demeaned plus/minus	Long difference	
	(1)	(2)	(3)	(4)	(5)
ln(lights/area)	0.281***	0.193***		0.321***	0.304***
	(0.031)	(0.037)		(0.037)	(0.037)
$\lvert + \Delta$ ln(lights/area)$\rvert$			0.273***		
			(0.039)		
$\lvert - \Delta$ ln(lights/area)$\rvert$			−0.288***		
			(0.056)		
ln(top-coded + 1)					0.020
					(0.015)
ln(unlit + 1)					−0.010
					(0.033)
Time effects	Yes	Yes	In demean	No	No
Country effects	Yes	Yes	In demean	No	No
Observations	3016	3016	3016	170	170
Countries	188	188	188	170	170
(Within) R^2	0.770	0.906	0.214	0.276	0.284

* p <0.1, ** p <0.05, *** p <0.01
Clustered (country) standard errors in parentheses.

which the calibration rests. The first row allows the elasticity to differ for countries with good data. The differences, 0.050, −0.005, and 0.092, are small and insignificant, as are the original 0.042, −0.014, and 0.096, which supports imposing a common relationship between lights and income on both groups. The second row regresses the squared residuals from Panel A on an indicator for good data. Since measurement error in GDP leaves the slope unchanged but adds to the residual variance, a negative coefficient indicates less noise in the GDP of good data countries. The coefficients, −0.0055, −0.0017, and −0.0281, match the original −0.0054, −0.0017, and −0.0292 in magnitude and significance. I do not report the Breusch-Pagan tests of the original table, which test the same hypothesis as the second row.

Table 3: Results for World Bank-Rated Countries

	Fixed effects	Country time trend	Long difference
Panel A			
ln(lights/area)	0.310***	0.280***	0.328***
	(0.038)	(0.045)	(0.046)
Constant			0.365***
			(0.028)
Observations	1954	1954	113
Countries	118	118	113
(Within) R^2	0.781	0.904	0.297
Panel B			
Difference in ψ for good data countries	0.050	−0.005	0.092
	(0.063)	(0.066)	(0.091)
Squared residuals on good data dummy	−0.005***	−0.002*	−0.028
	(0.002)	(0.001)	(0.018)

* p <0.1, ** p <0.05, *** p <0.01

Table 4 solves the measurement-error model for the five values of ϕ_g considered in the original Table 5. Because the signal share for bad data countries depends only on GDP, which matches the authors' data, ϕ_b reproduces the published values exactly. For the authors' preferred case of $\phi_g = 0.9$, the implied structural elasticity is $\beta = 1.134$, against 1.149, and the variance components are $\sigma_y^2 = 0.054$, $\sigma_{z,g}^2 = 0.006$, $\sigma_{z,b}^2 = 0.037$, and $\sigma_x^2 = 0.117$, against 0.054, 0.006, 0.037, and 0.128 in the original. The optimal weights on measured GDP are 0.850 for good data countries and 0.479 for bad data countries, against 0.852 and 0.484. As ϕ_g falls, my weights fall faster than the published ones, reaching 0.224 for bad data countries at $\phi_g = 0.6$, against 0.263. The source of this divergence is discussed in the next section.

Table 5 applies the weights for $\phi_g = 0.9$ to the thirty bad data countries, reproducing the original Table 6. For each country, the fitted value is the long-difference growth rate predicted from lights, the composite estimate places a weight of 0.48 on official growth and the remainder on the fitted value, and all growth rates are annualized as $100(e^{\Delta/13} - 1)$,

Table 4: Calibrating the Statistical Model

Signal to total variance of measured income			Weight for measured income growth in calculation of true growth	
Good data countries: ϕ_g	Bad data countries: ϕ_b	Structural effect of true income growth on lights growth β	Good data countries: λ_g	Bad data countries: λ_b
1	0.660	1.021	1.000	0.563
0.9	0.594	1.134	0.850	0.479
0.8	0.528	1.276	0.699	0.394
0.7	0.462	1.458	0.549	0.309
0.6	0.396	1.701	0.398	0.224

30 bad data countries, 83 good data countries.

where Δ is the log change between the two endpoints. The results match the original closely. For Myanmar, official growth of 10.02 percent per year and lights-based growth of 3.29 percent combine to 6.46 percent, a downward revision of 3.24 percent per year (as defined in Table 5), against 3.26, 6.48, and 3.22 in the original. At the other extreme, Burundi's official growth of -0.71 percent is revised upward to 1.13 percent. Across all thirty countries, the fitted and composite growth rates are within about 0.06 percentage points of the published values, and the ordering of countries by the size of the revision is the same except for a few adjacent pairs whose revisions are nearly identical.

Figure 1 reproduces the original Figure 7. Each point plots a bad data country's official growth against its growth predicted from lights, and countries above the 45-degree line are those for which lights suggest faster growth than the official figures. The dashed lines are iso-composite growth lines, each connecting the combinations of official and lights-based growth that yield the same composite estimate. Countries with the highest official growth, such as Myanmar and Angola, lie below the 45-degree line, and those with the lowest, such as Burundi and the Democratic Republic of Congo, lie above it. This is the pattern that measurement error in official statistics would produce, since a country's reported growth is extreme partly because of its error, and the composite estimates accordingly pull the extremes toward the center of the distribution.

Sources of Discrepancies

Most of the remaining differences can be traced to the construction of the country grid. The original analysis assigned pixels to countries using the authors' own rasterization of a cleaned version of the boundary polygons, which is included in their replication package (Henderson, Storeygard, and Weil 2012b) but depends on manual edits to the boundaries that are not documented. As a diagnostic, I recomputed the lights measure using the authors' grid while keeping every other part of my construction unchanged. The baseline elasticity falls from 0.281 to 0.278, within 0.001 of the published estimate, so the boundary construction accounts for nearly all of the difference in the lights coefficients. The remaining gap is consistent with the two data limitations described above: the missing F15 composite for 2008 and the use of Version 1 of the land area grid in place of the unreleased alpha version.

Table 5: Average Annual Growth Rates for Bad Data Countries, 1992/93-2005/06

Country	ISO	WDI GDP (LCU)	Fitted lights	Optimal combination of fitted lights and WDI GDP	Revision†
Myanmar	MMR	10.021	3.286	6.456	−3.240
Angola	AGO	6.991	3.885	5.360	−1.524
Nigeria	NGA	4.040	1.895	2.916	−1.080
Sudan	SDN	5.919	4.060	4.946	−0.919
Viet Nam	VNM	7.599	5.852	6.684	−0.850
Burkina Faso	BFA	5.801	4.460	5.100	−0.663
Benin	BEN	4.517	3.493	3.982	−0.512
Ghana	GHA	4.600	3.699	4.129	−0.450
Rwanda	RWA	3.055	2.239	2.629	−0.414
Algeria	DZA	3.287	2.833	3.050	−0.229
Oman	OMN	4.279	3.828	4.043	−0.226
Sierra Leone	SLE	3.045	2.717	2.874	−0.166
Cameroon	CMR	3.288	2.986	3.130	−0.153
Iran, Islamic Rep.	IRN	4.030	3.738	3.878	−0.146
Mali	MLI	5.079	4.808	4.937	−0.134
Niger	NER	3.481	3.240	3.356	−0.121
Gambia, The	GMB	3.797	3.771	3.783	−0.013
Liberia	LBR	6.747	7.013	6.885	0.130
Central African Republic	CAF	1.589	1.870	1.735	0.145
Mauritania	MRT	3.677	4.064	3.878	0.195
Eswatini	SWZ	3.417	3.930	3.684	0.259
Lebanon	LBN	3.849	4.447	4.161	0.300
Madagascar	MDG	2.743	3.367	3.068	0.316
Eritrea	ERI	3.508	4.920	4.241	0.709
Guinea-Bissau	GNB	−0.291	1.367	0.570	0.864
Congo, Rep.	COG	2.634	5.092	3.908	1.242
Haiti	HTI	−0.276	2.735	1.283	1.563
Cote d'Ivoire	CIV	1.815	4.919	3.422	1.578
Congo, Dem. Rep.	COD	−0.523	3.034	1.316	1.849
Burundi	BDI	−0.712	2.845	1.127	1.852

†Revision is the annualized gap between composite and WDI growth, $100\left(e^{(\Delta_c-\Delta_o)/13}-1\right)$, where Δ_c is the log change in composite and Δ_o is the log change in WDI GDP.

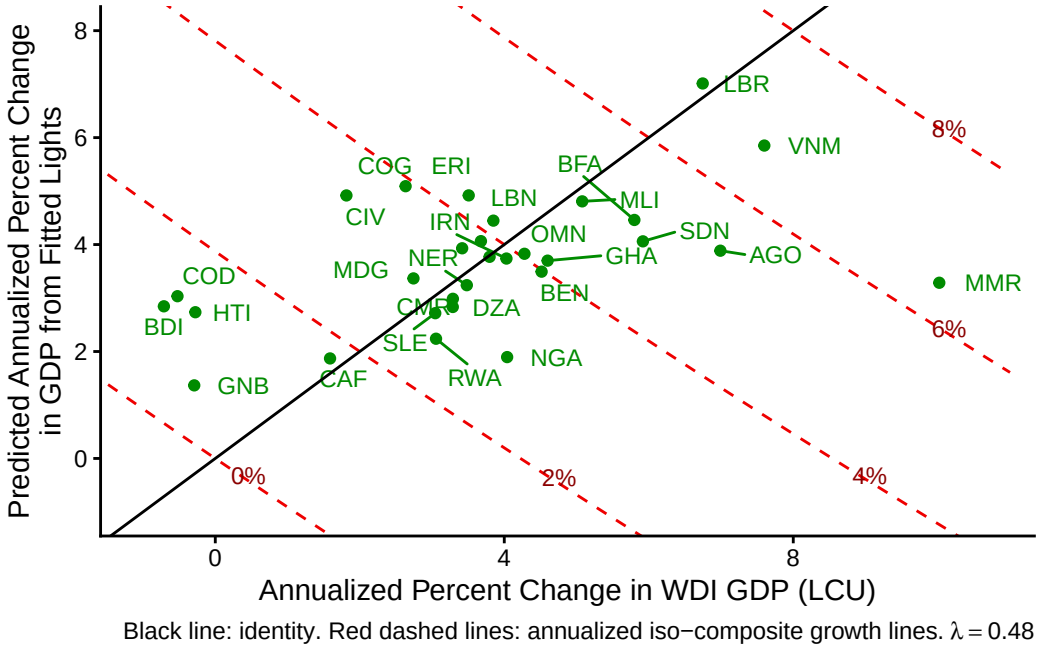


Figure 1: Growth in Fitted Lights vs. WDI GDP (LCU) for Bad Data Countries, 1992-2006

The divergence in Table 4 is attributable to a different source. The original HSW paper states that the variance of noise in lights is $\sigma_x^2 = \text{var}(x) - \beta^2\sigma_y^2 = \text{var}(x) - \beta\,\text{cov}(x,z)$, and this is the formula I use in Table 4. However, the authors' replication code diverges from the stated model and instead computes $\sigma_x^2 = \text{var}(x) - \text{cov}(x,z)$.[2] The variance of lights noise reported in the original for $\phi_g = 0.9$, 0.128, matches the code's formula, and recomputing the weights for bad data countries with that formula gives 0.485 at $\phi_g = 0.9$ and 0.264 at $\phi_g = 0.6$, close to the published 0.484 and 0.263. Because the implied β is close to one at the preferred value of ϕ_g, the two formulas give nearly the same weights there, and the composite estimates in Table 5 are essentially unaffected. The difference grows as β rises at lower values of ϕ_g.

Conclusion

In sum, I have successfully reproduced the central results of Henderson, Storeygard, and Weil (2012a) using public data sources. The elasticity of measured GDP with respect to lights is close to 0.3 across specifications. I also confirm that lights are nearly equally effective at tracking expansions and contractions, that the relationship between lights and income is the same for countries with good and bad statistics, and that the GDP of countries with bad statistics is measurably noisier.

[2] In `lightspaper_replication.do` in the replication package (Henderson, Storeygard, and Weil 2012b), line 470 computes `sigmasqx = varx - cov`, which equals $\text{var}(x) - \text{cov}(x,z) = \text{var}(x) - \beta\sigma_y^2$ rather than $\text{var}(x) - \beta^2\sigma_y^2$.

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